FRANK DUMOLUHLE NCUBE

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EDUCATION

Livingstone College | B.S. Computer Information Sciences **Expected December 2029**
Salisbury, NC | GPA: 4.0/4.0 | International Presidential Scholar
Relevant coursework: Programming Fundamentals, Computer Information Systems, Networks & Telecommunications, College Mathematics

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS
Web & data: Next.js, React, Node.js, Express, PostgreSQL, Supabase, REST APIs, Vercel, Tailwind CSS
AI & engineering: LLM API integration, retrieval (TF-IDF), prompt engineering, Git/GitHub, Vitest, AWS fundamentals

SELECTED TECHNICAL PROJECTS

ShopAccess AI | AI shopping assistant with permission-based checkout **2026**
PayPal AI Hackathon (solo entry) | Claude Code-assisted, Codex-reviewed | Next.js, TypeScript, Zod, Vitest, Chrome MV3
- Led AI-assisted development of a bilingual (English/Spanish) laptop-shopping prototype that turns natural-language needs into shopper-confirmed requirements and server-validated recommendations.
- Directed an architecture separating advisory AI output from deterministic server checks: AI-hardware compatibility shows “not verified” without evidence, and checkout requires a revocable, item-specific purchase permission with quantity, total, and expiry limits.
- Scoped a paired Chrome extension side panel (cannot confirm or pay) and an optional PayPal Sandbox flow; default checkout is simulated. 2,568 automated tests pass locally.

EtenMall | Multi-tenant e-commerce platform (in development) **2026–Present**
Next.js, TypeScript, Supabase, PostgreSQL, Vercel
- Founder; designed and directed AI-assisted development of a tenant-isolated retail platform (storefronts, branch inventory, variants, staff roles, guest checkout) with PostgreSQL row-level security and tested privilege boundaries across 83 migrations.
- Developed an AI-assisted workflow with written specifications and approval gates; directed a PostgreSQL deadlock fix (deterministic lock ordering) and staged, flag-gated releases. Paynow test mode only; 4,412 unit tests pass locally.

Triage360 | AI support triage system **2026**
React, Express, PostgreSQL, TypeScript, Python, Claude API
- Built ticket classification, document retrieval, and grounded AI responses with human escalation for six risk categories.
- Added ticket history and statistics; built a separate Python prototype exploring TF-IDF and semantic retrieval.

Horizon Desk | AI banking relationship prototype **2026**
SteelHacks XIII (team of three) | Next.js, TypeScript, Claude 3.5 Sonnet, ElevenLabs, Vitest
- Co-developed the frontend and supporting backend for a synthetic-data prototype that applies deterministic eligibility checks before AI-assisted outreach and routes credit opportunities to human review.

EXPERIENCE

Livingstone College Academic Support | Math & science tutor **2026–Present**
- Tutor 10+ students in individual and small-group sessions, adapting explanations and practice to learning needs.

Frank Wi-Fi Hotspot Business | Founder & operator **2024–Present**
- Operate a Starlink-powered voucher internet service in Gweru, Zimbabwe; coordinate approximately 30 people across sales, installation, marketing, and maintenance, and troubleshoot network issues remotely.

CERTIFICATIONS, TRAINING & LEADERSHIP

AWS Academy Graduate: Generative AI Foundations (2026); Cloud Foundations
Datafrik Advanced AI & MLOps Bootcamp — In progress (11 weeks): model evaluation, neural networks, NLP, API deployment, Docker, CI/CD, monitoring
Leadership: HBCUniverse Campus Ambassador; HBCU Heroes participant